



Institute of Strategic and International Studies
Economics, Trade
and Regional Integration

Novel AI technologies and the future of work in Malaysia

3 December 2024

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and Regional Integration**

About Economics at ISIS Malaysia

ISIS Malaysia has for over four decades conducted evidence-based research on economics, foreign policy, social policy, and climate policy. The Economics, Trade, and Regional Integration programme at ISIS Malaysia focuses on analysis of emerging economic policy issues in Malaysia and the region.

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Overview

- ▶ **Motivation**
- ▶ **Methods**
- ▶ **Early results**





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► Why do we care about new technologies and work?

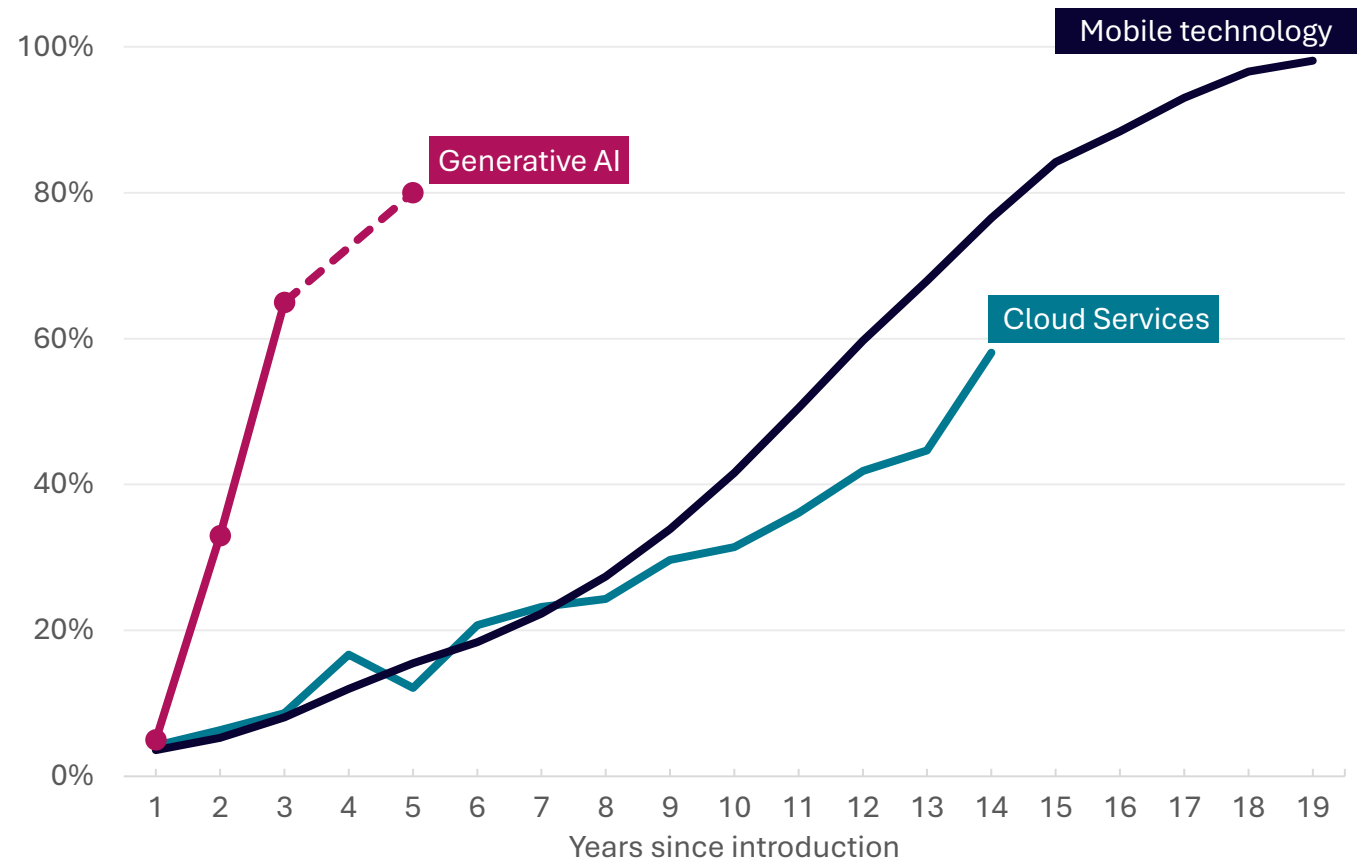
- **Technology is the main driver of change in the world of work.** Since early human history, it has repeatedly changed what work is and how humans perform it. (e.g. first agriculture revolution¹, steam engine)
- **Each wave of technological change is seen as progress.** But progress is not universal, technological change often creates significant negative consequences for societies. (i.e. industrial revolution², robotisation³, ICT⁴)
- **The pace of technological change is increasing**, and its consequences on work is becoming more widespread. How this affects societal wellbeing is a policy and political choice.⁵

Source: ¹Chu et al. (2020), ²Frey (2024),
³Graetz and Michaels (2018), ⁴Acemolgu and Restrepo (2019), Katz and Goldin (2007), Autor (2020). ⁵Acemoglu (2022), Bastani (2019)

► What makes novel AI technologies different from past waves?

Source: ISIS Malaysia staff estimates using data from McKinsey & Co, Gartner, OECD, ITU
Note: Adoption of cloud services are OECD averages

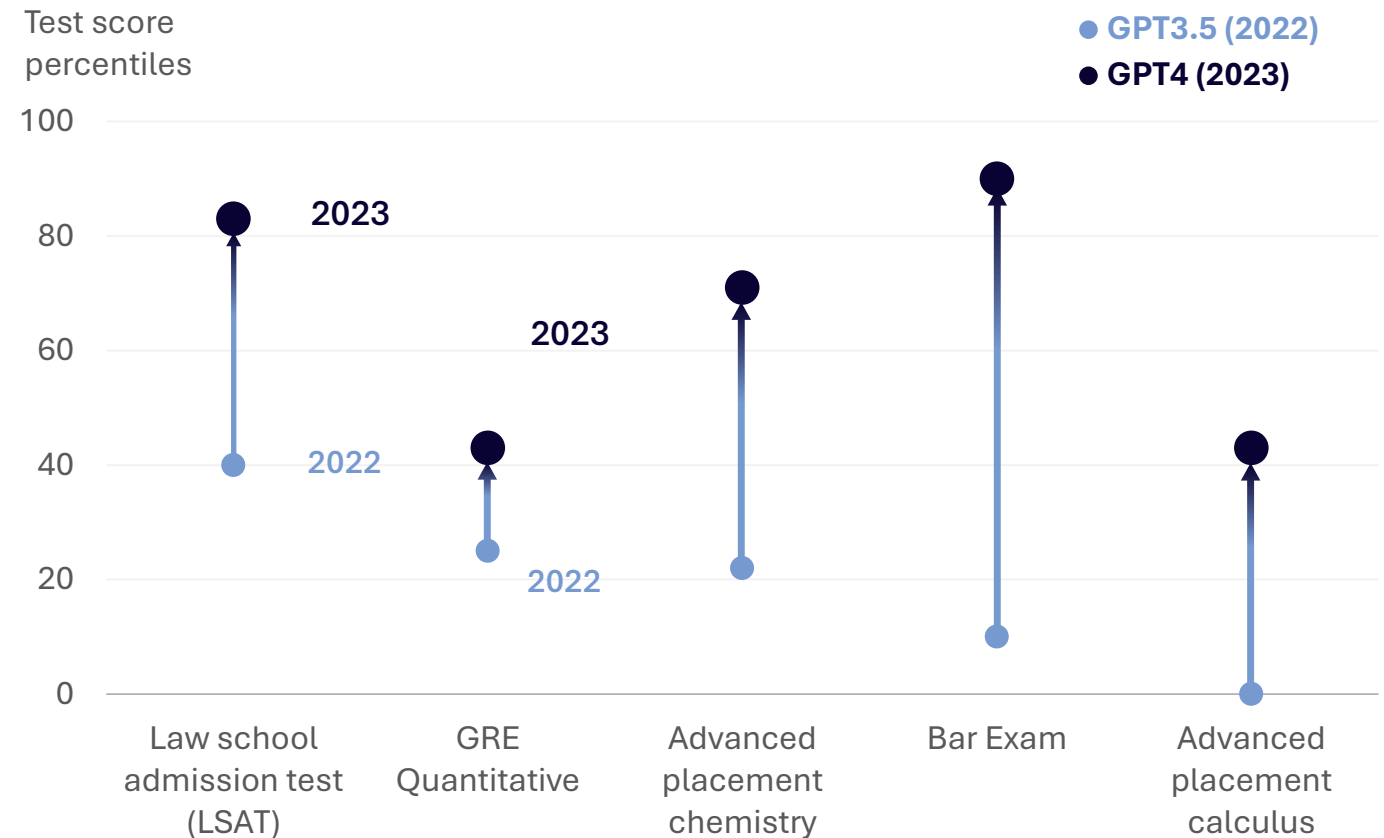
- **AI diffusion has been far more rapid than past tech.**
Adoption of generative AI technologies have been about 5x faster than mobile phones



► What makes novel AI technologies different from past waves?

► Pace of AI development has also far quicker.

Generative AI capabilities are converging to frontier of human cognitive abilities



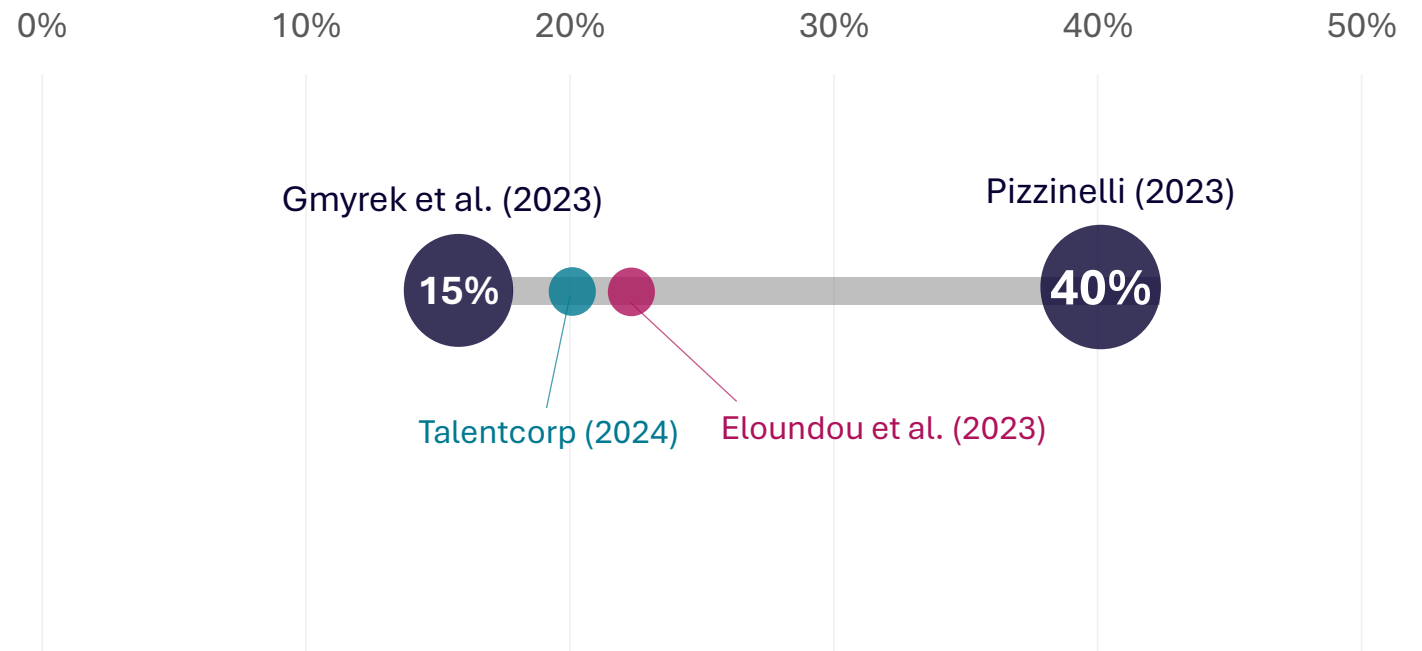
Source: Adapted from Eloundou et al. (2023)

Note: Estimates refer to the 2023 version of GPT4 and 2022 version of GPT3.

► What does the recent literature say on the job impacts of generative AI?

► AI exposure is relatively high, but not all of those exposed will be displaced.

% share of jobs highly exposed or impacted by generative AI



Source: Authors' illustrations based on Gmyrek et al. (2023), Talentcorp (2024), Eloundou et al. (2023), Pizzinelli (2023).

Note: Gmyrek et al. (2023): % of global jobs, Talentcorp (2024): % of Malaysian workers, Eloundou et al. (2023): % of US workers, Pizzinelli (2023): % of jobs in emerging market countries.



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- ▶ **Methods**
- ▶ Early results



► What questions are we trying to answer?

1

How does emerging generative AI technologies affect jobs, and who does it affect most?

(across occupations, demographics, geography)

2

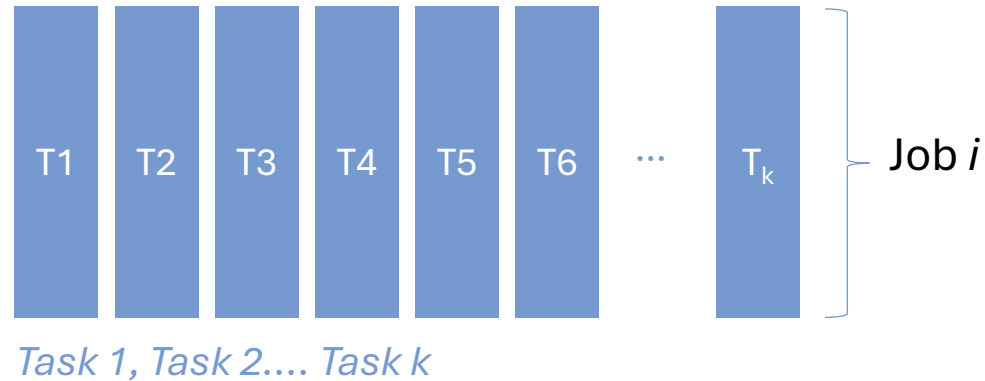
What types of policies are needed to maximise the benefits of technology diffusion while limiting its negative impacts?

(education, skills, social protection)

► Start by building a conceptual framework of jobs

► What is a job?

Jobs as a collection of discrete tasks



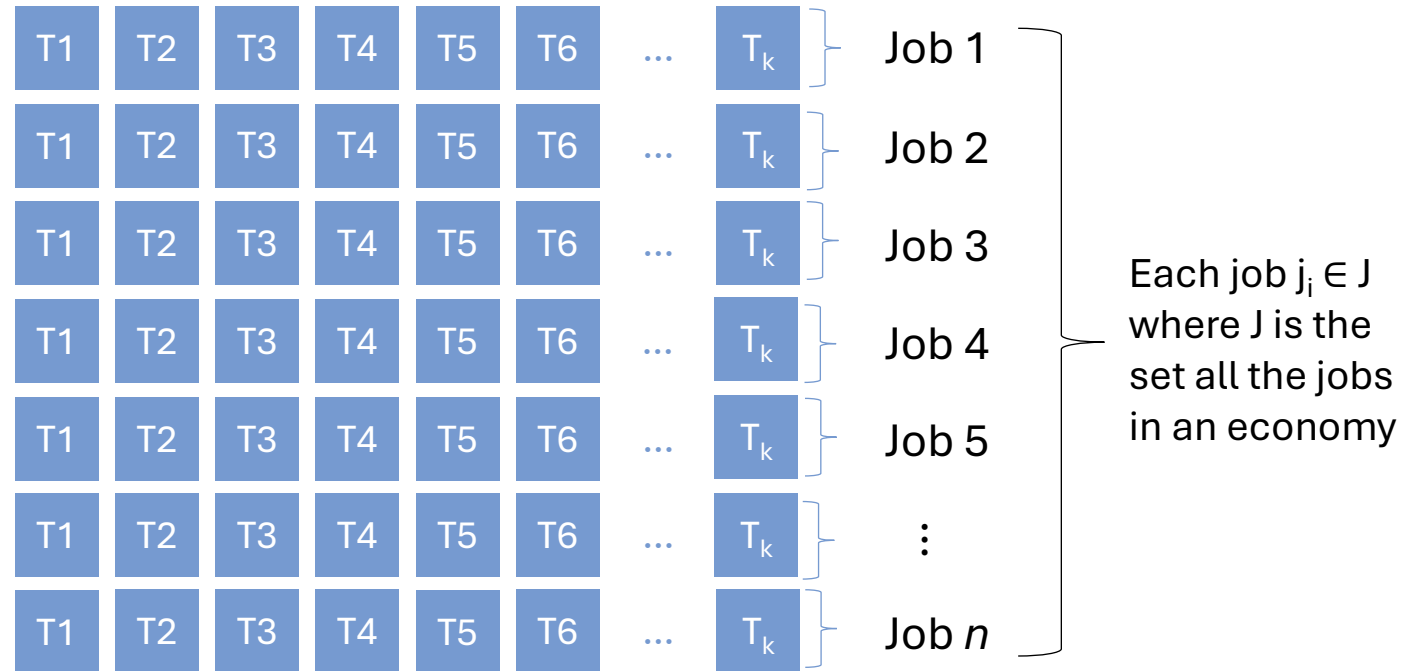
Define each job j_i as containing a set of tasks T_i such that $T_i = \{t_1, t_2, \dots, t_k\}$

Source: Authors' illustrations based on Cheng and Chong (2024).

► Start by building a conceptual framework of jobs

► What is a job?

Jobs as a collection of discrete tasks

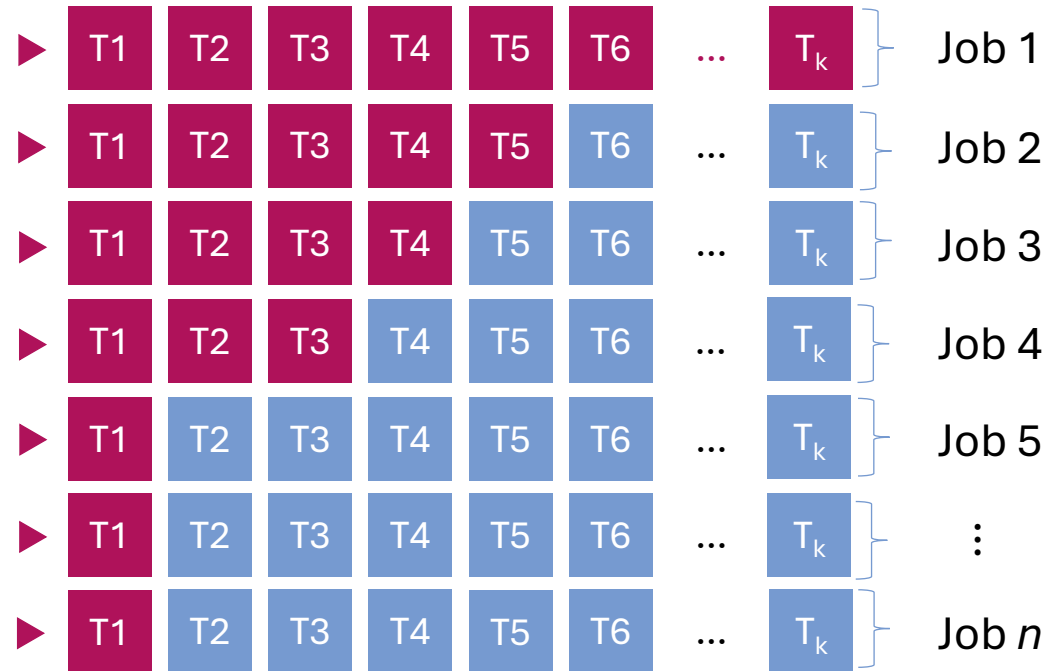


Source: Authors' illustrations based on Cheng and Chong (2024).

► How does technology affect jobs in our framework?

► Technology affects jobs by affecting its tasks.

Different tasks are affected differently



Source: Authors' illustrations based on Cheng and Chong (2024).



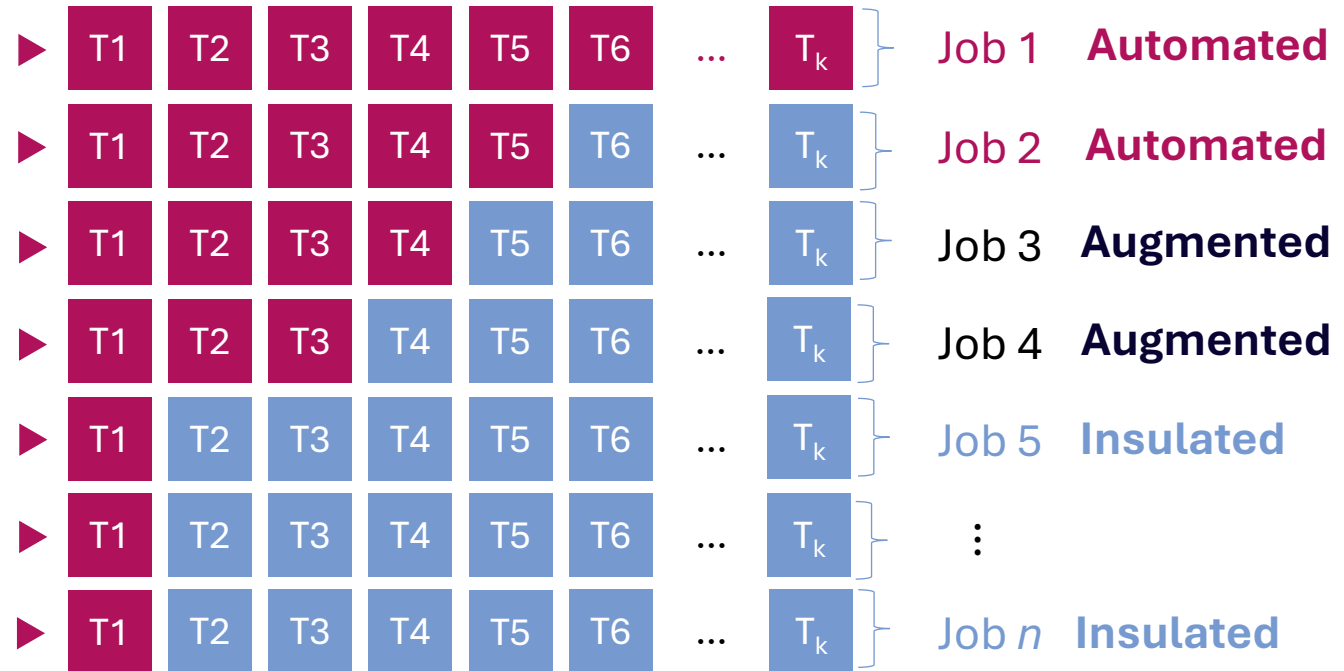
Automatable by
generative AI



Not deemed automatable
by generative AI

► How do we aggregate automation intensity of tasks to classify job impacts?

► We can classify job impacts based on the automation intensity of its task mix.



Source: Authors' illustrations based on Cheng and Chong (2024).

■ Automatable by generative AI

■ Not deemed automatable by generative AI

► How do we aggregate automation intensity of tasks to classify job impacts?

► Setting this out formally...

- For each job-task, define automatability score $A(t_{ik})$, where $0 < A(t_{ik}) < 1$
- Next, count number of tasks in each job i that exceeds a pre-defined automatability threshold θ .

$$P_{ik}(\theta) = 1\{A(t_{ik}) > \theta\}$$
- Then, we can define an ‘automation intensity’ A_i for each job i , $A_i = k_\theta/k$, which measures the proportion of tasks within a job that is susceptible to significant automation by generative AI

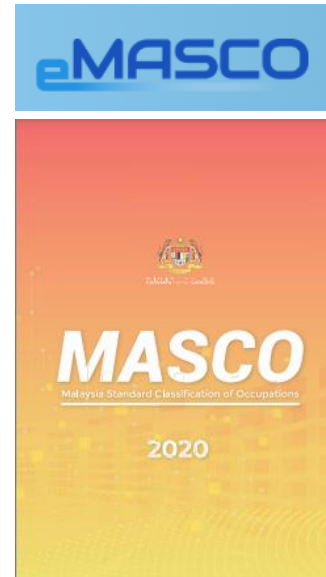
Then we can classify jobs as...

- **Automatable:** $A_i > \alpha_{\text{auto}}$
- **Augmentable:** $\alpha_{\text{aug}} < A_i < \alpha_{\text{auto}}$
- **Insulated:** $A_i < \alpha_{\text{aug}}$

Source: Cheng and Chong (2024).

► Bringing the conceptual framework to the real world

► Retrieving the set of jobs and tasks in an economy Using the Malaysian Standard Classification of Occupations (MASCO)



Classification Structure

- > 1 Managers
- ✓ 2 Professionals
 - > 21 Science And Engineering Professionals (8)
 - > 22 Health Professionals (6)
 - > 23 Teaching Professionals (9)
 - ✓ 24 Business And Administration Professionals (3)
 - > 241 Finance Professionals (3)
 - ✓ 242 Administration Professionals (6)
 - > 2421 Management And Organisation Analysts (16)
 - ✓ 2422 Policy Administration Professionals (2)
 - 2422-01 Administrative and Diplomatic Officer Grade M41
 - 2422-02 Policy Analyst
 - > 2423 Personnel And Career Professionals (14)
 - > 2424 Training And Staff Development Professionals (8)
 - > 2425 Administrative Professionals (15)
 - > 2426 Research And Development Professionals (21)
 - > 243 Sales, Marketing And Public Relations Professionals (4)
 - > 25 Information and Communications Technology Professionals (4)
 - > 26 Legal Professionals (1)
 - > 27 Hospitality and Related Services Professionals (1)
 - > 28 Social And Cultural Professionals (4)
 - > 29 Regulatory Government Professionals (1)
- > 3 Technicians and Associate Professionals
- > 4 Clerical Support Workers
- > 5 Service and Sales Workers
- > 6 Skilled Agricultural, Forestry, Livestock and Fishery Workers
- > 7 Craft and Related Trades Workers
- > 8 Plant and Machine Operators and Assemblers

492 jobs (4-digit level) and
3958 jobs (6-digit level)

Source: MASCO 2020
<https://emasco.mohr.gov.my/masco/>

► Bringing the conceptual framework to the real world

► Retrieving the set of jobs and tasks in an economy

Using the Malaysian Standard Classification of Occupations (MASCO)

The logo for eMASCO, featuring the text "eMASCO" in white on a blue rectangular background.

Political Scientist

MASCO Code : 2823-04

Task Includes

- Researching political subjects, such as the country's political system and foreign relations
- Collecting and analyzing data from sources such as public opinion surveys
- Developing and testing political theories
- Evaluating the effects of policies and laws on government, businesses, and people
- Improving understanding and coordinating work execution
- Performing related tasks

The set of tasks $T_i = \{t_1, t_2, \dots, t_6\}$

~3,500 tasks at the 4-digit level, ~20,000 tasks at the 6-digit level

► Assigning automatability scores to the job-task list

► Using sequential API calls to GPT-4o

Following Gmyrek et al. (2023) and Eloundou et al. (2023), we generate automatability scores for tasks using a modified training prompt

```
{“role”: “system”, “content”: “You are a skills and AI specialist. ” + “You will provide a score of potential automation with GPT technology for a given task. Follow instructions closely.”},
```

```
{“role”: “user”, “content”: “Look at this job task: ” + str(Tasks_GPT) + “It is related to MASCO code: ” + “Provide a score of potential automation of this task with Generative AI technology, given that the job is located in Malaysia: ” + “The score should range 0-1. Provide a score in one line, and a justification in next line. Do not provide any other commentary, only the score and justification. ” + “Do not give any ranges just one score for each task.”}
```

Source: Authors' illustrations using Gmyrek et al. (2023), Eloundou et al. (2023).



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► Early results at the 4-digit level:
Automatability scores for individual tasks

Source: Authors' estimates. Early results - do not cite.

- **We generate automatability scores for each job-task**
 Scores range from 0 (least automatable) to 1 (most automatable)

Child care worker
 MASCO Code: 5311

Task	$A(t_{ik})$
Assisting children to bath, dress and feed themselves	0.05
Taking children to and from school or outdoors for recreation	0.1
Playing games with children, or entertaining them by reading or story-telling	0.3
Looking after the welfare and safety of children in the absence of their parents	0.1
Maintaining order in children's bedrooms and playrooms	0.1
Taking care of school children on excursions, museum visits and similar outings	0.1

Automation intensity low = Insulated

► Early results at the 4-digit level:
Automatability scores for individual tasks

Source: Authors' estimates. Early results - do not cite.

- **We generate automatability scores for each job-task**
Scores range from 0 (least automatable) to 1 (most automatable)

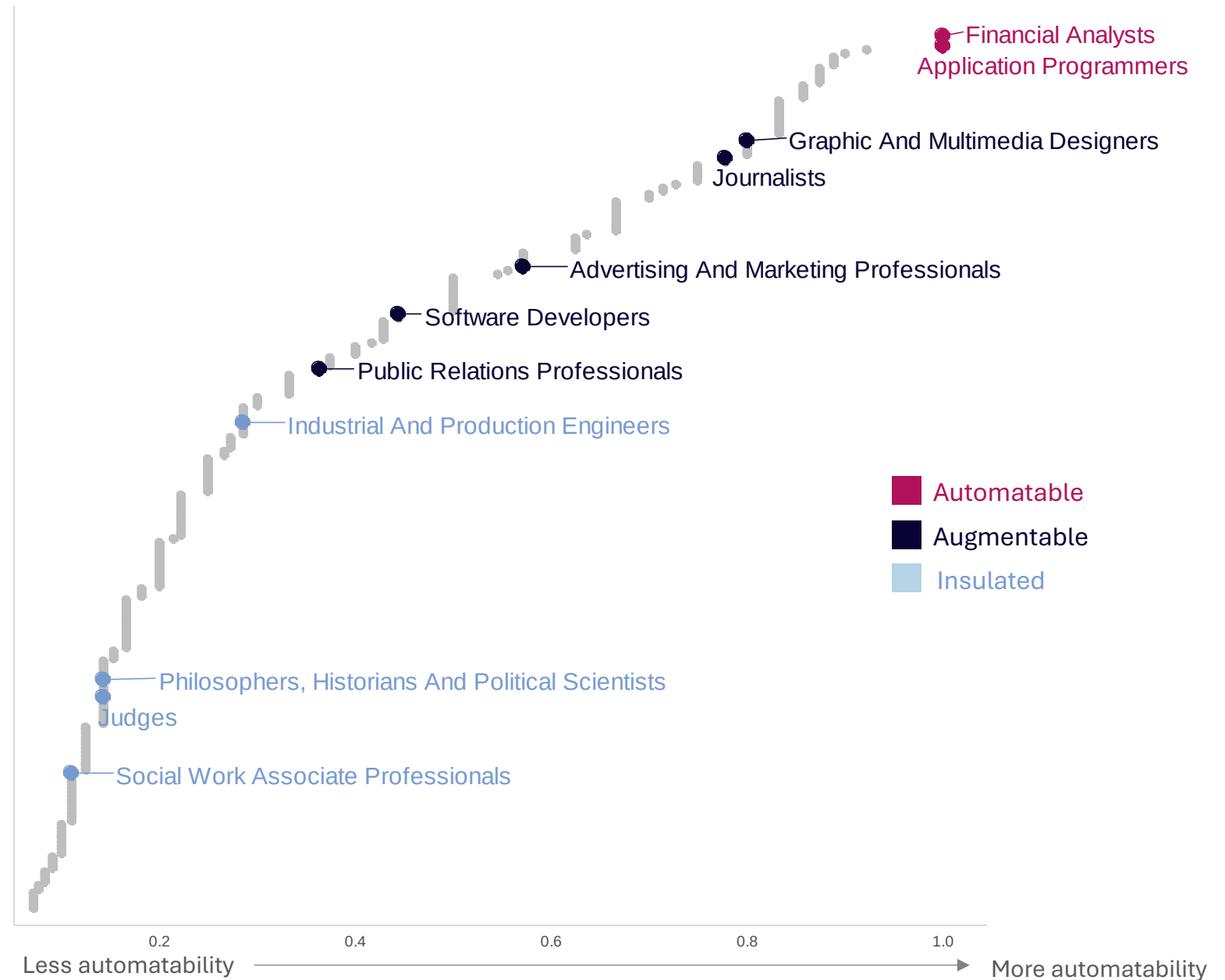
Writers (including copyeditors, proofreaders)
MASCO Code: 2831

Task	$A(t_{ik})$
Writing literary works for publication or dramatic performance	0.6
Writing scripts and continuities and preparing programmes for stage, film, radio and television productions	0.65
Appraising and writing about literary, musical and other works of art and artistic performances	0.65
Writing advertising copy promoting a particular product or service	0.85
Selecting, assembling and preparing publicity materials about business or other organisations for issue through the press, radio, television and other media	0.75
Designing and writing brochures, handbooks and similar technical publications	0.75

Automation intensity high = automatable

► Early results at the 4-digit level:
Which occupations are most affected?

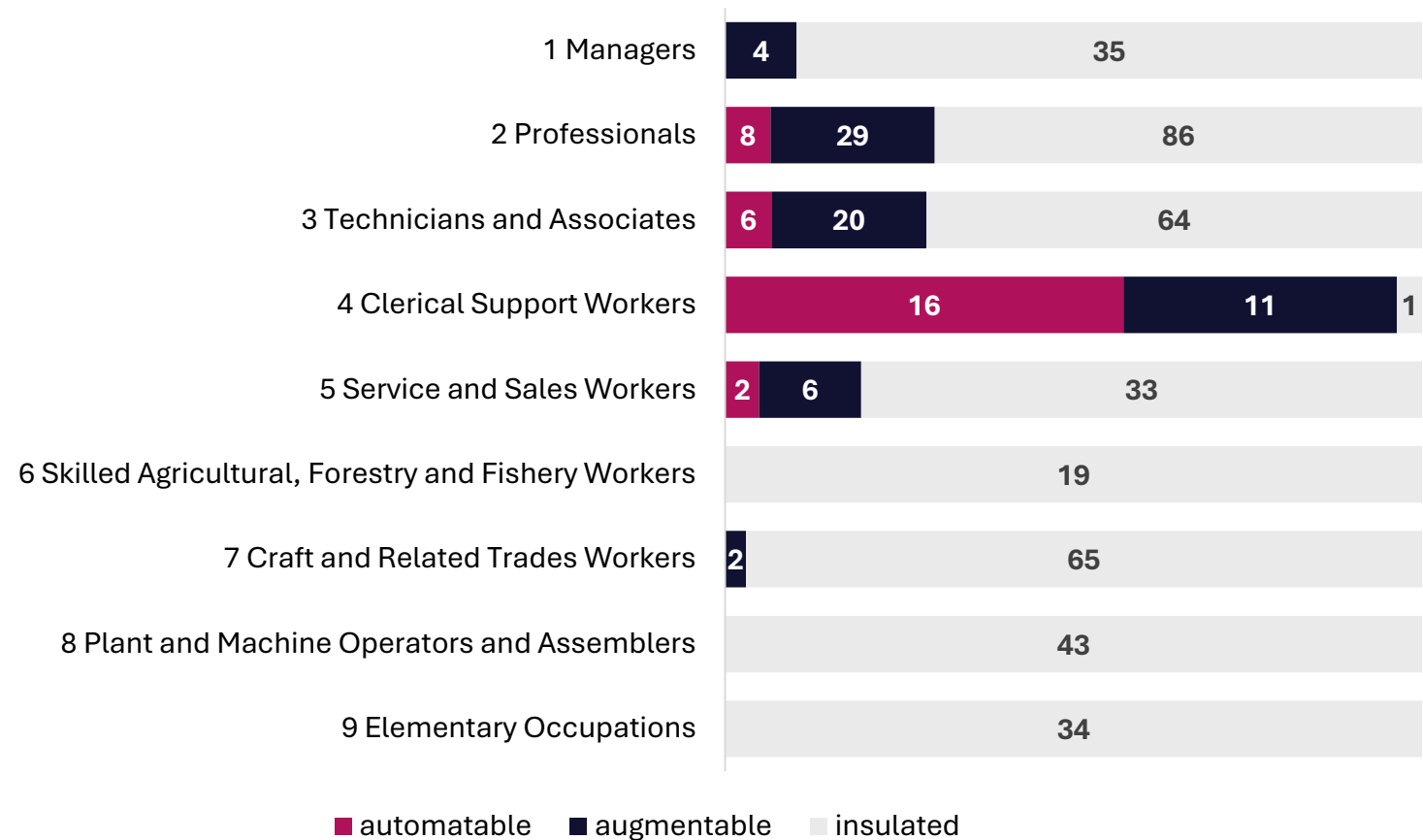
Source: Authors' estimates. Early results - do not cite.



► Early results at the 1-digit level:
Which broad sector is most affected?

► **Clerical support occupations are most exposed to automation by generative AI.**

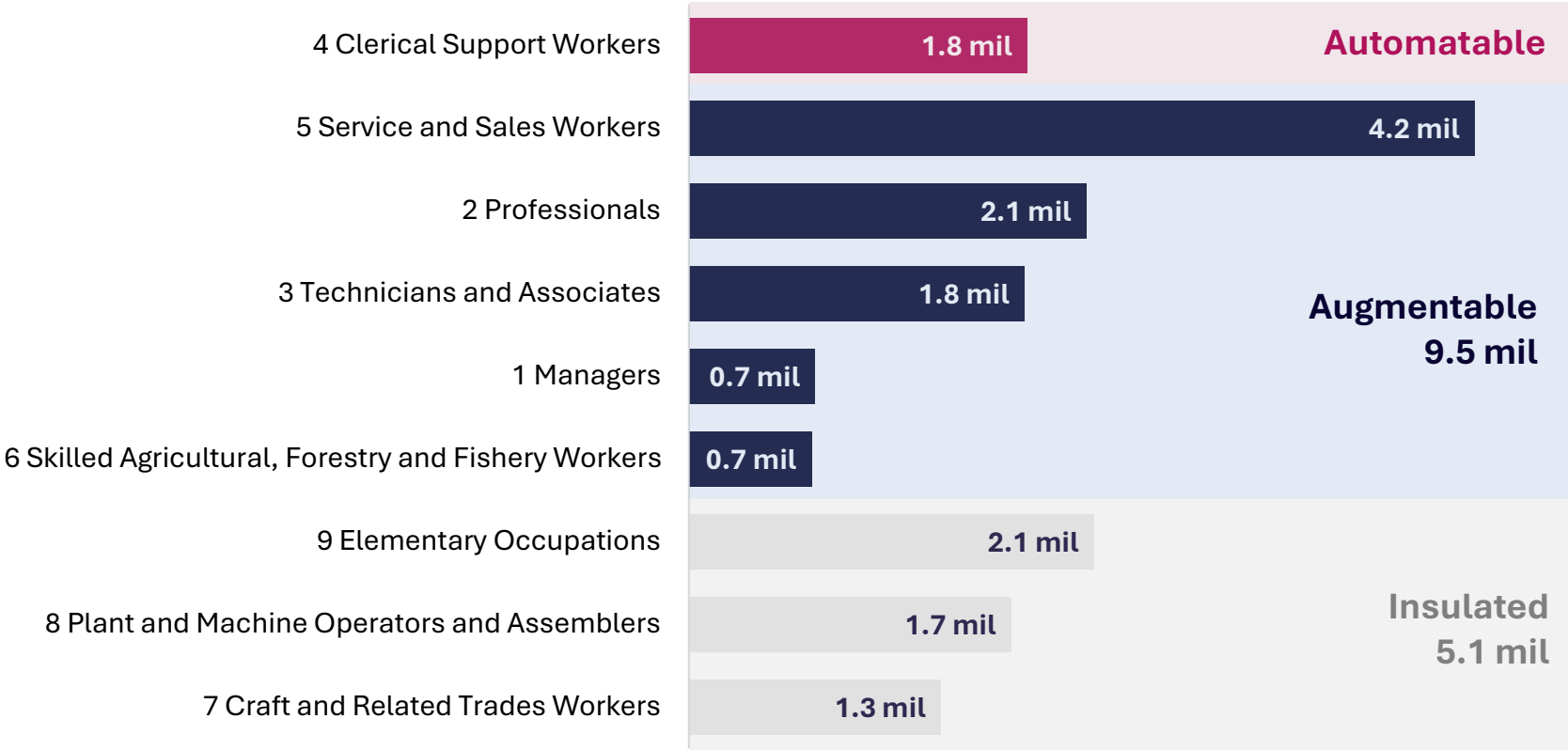
No. of occupations exposed to generative AI



Source: Authors' estimates. Early results - do not cite.

► Early results at the 1-digit level:
Which occupations are most affected?

► **Most Malaysian workers are in occupations that are augmentable by generative AI.**
Number of employees and their AI exposure



Source: Authors' estimates. Early results - do not cite.

► **Next steps in this research project**

- **Linking our estimates with labour force survey data.**
Mapping our AI automatability scores onto detailed labour force and demographic microdata will help us to analyse the socio-demographic, geographic, gendered impacts of generative AI.
- **Extensions to MASCO 6-digit level.**
Running the analysis for the 6-digit level will give us a better understanding of how specific occupations are affected.
- **Expanding to more countries across the Asia Pacific.**
We are planning to expand the country scope of this work to more countries in 2025, particularly in ASEAN.
- **Exploring other margins of adjustment.**
We want to investigate other related areas including macroeconomic and productivity impacts, new task creation etc.

Source: Authors' estimates. Early results - do not cite.

► **Key takeaways and concluding thoughts**

► **The pace of technological change is increasing.**

Gen AI is uniquely disruptive compared to past waves due to its speed of development and rapid diffusion.

► **Early evidence suggest relatively high percentage of exposure, though not all likely displaced.**

Skills not automatable likely related to human creativity, socio-emotional intelligence, and physical labour.

► **Technological change has not (yet) eliminated human labour.**

But we are getting pretty close. Current financial and economic (capitalist) incentives tend to favour labour-replacing technologies over labour-enabling technologies.



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Thank you.

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